Ogbonna Arinze

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SUMMARY

Senior Frontend Engineer with 8 years of professional experience building scalable, data-intensive web applications using **React**, **TypeScript**, and **Node.js** across fintech, healthcare, and civic tech environments. Proven track record of architecting **design systems** and reusable UI component libraries, optimizing frontend performance in distributed service architectures, and delivering complex data visualization interfaces. Experienced in fintech product development including real-time transaction processing, payment module integrations, and financial data dashboards.

CORE COMPETENCIES

- **Frontend Architecture:** React, TypeScript, Next.js, scalable component systems, design systems (Radix UI, Storybook), monorepos (Nx), and SSR/SSG.
- **UI/UX Implementation:** Design system development, Tailwind CSS, Ant Design, Figma-to-code workflows, accessibility (WCAG/ARIA), and responsive layouts.
- **Node.js & Backend:** Node.js services, NestJS, REST APIs, GraphQL, JWT authentication, role-based access control, and MongoDB/PostgreSQL.
- **Data Visualization & Real-Time Systems:** react-chartjs, Recharts, RxJS, EventSource API, SSE streaming, and high-volume transaction processing.
- **Testing & Quality:** Playwright, Jest, Enzyme, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA pipelines.
- **Cloud & Infrastructure:** AWS (CloudFront, S3, Lambda), Docker, GitHub Actions, CI/CD pipelines, and Kubernetes awareness.
- **Performance & Observability:** Core Web Vitals, code splitting, bundle analysis (Webpack, Rollup), Lighthouse, React Profiler, and CDN strategies.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and cross-functional collaboration with UX/UI teams.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer (React / TypeScript) — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension built with **TypeScript**, optimizing workflows for clinical site integrations and enhancing adoption across distributed services.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams, serving as the shared UI library for all frontend applications.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using OpenAI GPT models, LangChain, Vercel AI SDK, and Chroma vector database, enabling users to query their data in real time via a conversational interface.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools profiling.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews and pair programming.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface with **TypeScript**, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and improved release confidence.

- Utilized **Storybook** for documenting React components, creating a living component library that streamlined change management and ensured consistent design implementation across teams.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the frontend codebase.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, processing thousands to millions of daily financial transactions and boosting settlement rates by **80%**, handling complex data flows across distributed banking services.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests to distributed backend services.
- Built a real-time account reconciliation solution, boosting daily transaction reconciliation by **60%** through innovative data processing and management strategies in a high-throughput fintech environment.
- Adopted the native **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster updates and a more responsive interface in the transaction monitoring dashboard.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design** UI components.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models, delivering testable, maintainable, and well-documented backend services.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS** production servers.
- Mentored two interns through frontend and backend projects, providing technical guidance, code reviews, and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** with **TypeScript** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with 22 interactive chart types (**Recharts**), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring 22 interactive data visualization chart types auto-generated from AI responses, dark mode, URL state management with nuqs, and responsive mobile layouts.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, **PostgreSQL 16** with **Prisma v7**, phone OTP authentication, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*